

GRAPE: Graph-based KV Cache Selection for Efficient Long-context LLM Inference

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Abstract

As the demand for long-context large language models (LLMs) grows, managing the increasing memory and computational overhead during inference has become a significant challenge, particularly due to the expanding key-value (KV) cache size. Existing KV cache selection algorithms, while effective in reducing memory requirements, suffer from key limitations: (1) they either rely solely on static strategies or heuristic methods that lack adaptability and query awareness or (2) they fail to effectively leverage the similarity of token importance across consecutive layers, leading to inefficiencies and degraded performance in long-context inference. To address these challenges, we¹ propose **GRAPE**, a graph-based KV cache selection algorithm that integrates historical information from previous decoding steps and vertical information from prior layers. GRAPE introduces the Key Neighbor Graph (KNG) to cluster keys during the prefill stage, leveraging attention weight history. In the decode stage, GRAPE performs sparse attention by inheriting tokens from prior layers and expanding candidate tokens using KNG, followed by fine-grained token selection through quantized dot-product scoring. Evaluation on the RULER benchmark demonstrates that GRAPE achieves a token hit rate of up to 89% and task accuracy of 74.40% with only 16 tokens in a 4K context, significantly surpassing state-of-the-art methods like Quest. GRAPE can reduce up to 250× less memory footprint of KV cache during LLM inference.

1 Introduction

Large Language Models (LLMs) have seen continuous advancements, achieving superior performance across a wide range of tasks while also extending their capabilities to handle increasingly longer contexts. Long-context LLMs, such as GPT-4 (OpenAI

et al., 2024), Llama-3 (Grattafiori et al., 2024), and Gemini (Team et al., 2024), can process the context window of million tokens. However, the larger context window substantially amplifies memory and computational demands, driven by the autoregressive generation of LLMs. To reduce the overhead, the key-value (KV) cache is used to select and store attention keys and values.

Inspired by the sparsity nature of the attention mechanism, KV cache selection is applied to enhance long-context inference efficiency. However, most of the prior methods only depend on static strategies or heuristic methods that only leverage historical information from the current layer or vertical information from previous layers.

Static KV cache selection strategies, such as StreamingLLM (Xiao et al., 2023) and LM-Infinite (Han et al., 2023), face the limitation of forgetting information midway through processing and not being query-aware.

Vertical heuristic strategies, such as TidalDecode (Yang et al., 2024), which selects top-k tokens from specific layers as token sets, lack an effective updating mechanism to address shifts in important tokens across layers.

Historical heuristic methods, such as H2O (Zhang et al., 2024) and Quest (Tang et al., 2024), overlook the vertical information to enhance the efficiency and effectiveness. Among these methods, Quest (Tang et al., 2024) that uses page-quantized attention to select top token pages stands out as the state-of-the-art (SOTA) strategy. However, lacking vertical information, Quest requires scanning the entire KV cache and struggles with long-distance dependencies due to the continuous paging of candidates, highlighting room for improvement in both efficiency and effectiveness.

In this work, we observe the similarity between continuous layers in both important tokens and attention score distribution, as shown in Figure 1 and Figure 2. The attention similarity between continu-

¹* means equal contribution. Report link is here: <https://www.overleaf.com/read/vmjvcdhkkvznz#7cd35d>.

ous layers indicates that information from previous layers can be leveraged to improve KV cache selection.

With the observation, we propose GRAPE, a graph-based KV cache selection algorithm for efficient long-context LLM inference that maintains a key neighbor graph (KNG) using attention weight history from the current layer and depends on pivot tokens obtained from the previous layer. Combining both historical and vertical information, GRAPE improves the accuracy of selecting critical tokens. We benchmark the accuracy and efficiency of GRAPE on the RULER (Hsieh et al., 2024) dataset. Under the same budget of only 16 tokens in 4k context length, GRAPE achieves 74.40% average accuracy in 13 tasks of RULER compared with 58.20% in SOTA Quest (Tang et al., 2024) indicating the efficiency and effectiveness of GRAPE.

In summary, our contributions are as follows:

- We analyze the attention similarity between the consecutive layers, paving the road for developing more efficient KV cache selection methods. See Figure 2, Figure 1 and Section 4.2.
- We propose GRAPE, an efficient Graph-based KV Cache Selection algorithm that leverages the key neighbor graph to incorporate information from previous layers and former tokens. See Section 3.
- We evaluate GRAPE on the RULER dataset, where it achieves a high token hit rate (74-89%) and great task accuracy (74.40%) in a very small budget (e.g., 16 tokens), demonstrating strong performance in the RULER benchmark. See Section 4.

2 Preliminaries & Related Work

2.1 Problem Formulation

In the standard attention mechanism, the attention output o is computed as a weighted average of the value vectors, where the attention weights $\omega \in R^{1 \times n}$ are determined by taking the softmax over the query-key dot product scores:

$$\omega = \text{Softmax}\left\{\frac{qK^T}{\sqrt{d}}\right\} \quad o = \omega V \quad (1)$$

$q \in R^{1 \times d}$ denotes the current query vector, $K = [k_1, k_2, \dots, k_n] \in R^{n \times d}$ denotes the key vectors,

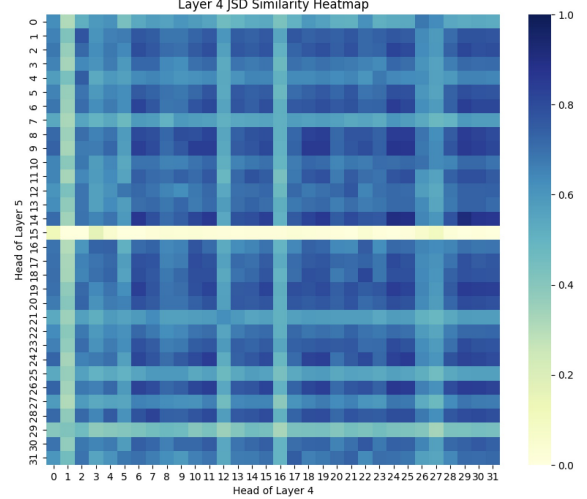


Figure 1: Attention JSD (Menéndez et al., 1997) similarity between all attention heads of layer 3 and layer 4 in Llama3-8B model with frequent word extraction(FWE) task.

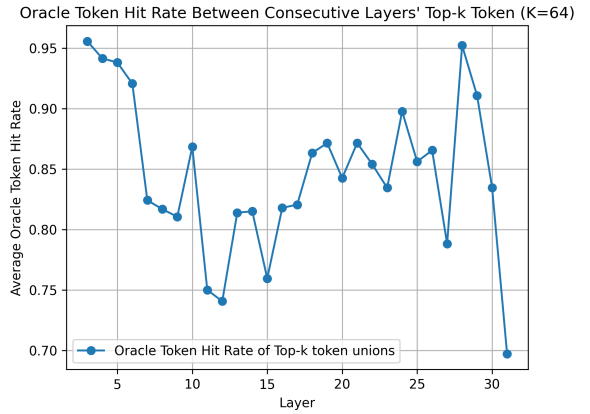


Figure 2: Top-64 important (a.k.a, oracle) token hit rate between two continuous layers in Llama3-8B. Each data point in layer L means the hit rate of top-64 tokens if we directly adopt top-64 tokens from layer $(L-1)$.

$Q = [q_1, q_2, \dots, q_n] \in R^{n \times d}$ denotes the value vectors, where d is the head dimension and n is the context length.

To minimize computational overhead, a set of important tokens $S = [s_1, s_2, \dots, s_b], b < n, s_i \in [n]$ is selected from KV cache. Then the attention output is computed only using the selected key and values:

$$\omega_S = \text{Softmax}\left\{\frac{qK_S^T}{d}\right\} \quad o_S = \omega_S V_S \quad (2)$$

2.2 Sparsity of Attention in LLMs

The sparsity of attention (Han et al., 2023; Xiao et al., 2023; Liu et al., 2024; Tang et al., 2024) is a common characteristic observed in LLM. Due to

the existence of attention sinks (Xiao et al., 2023), the attention weights are heavily concentrated on a small subset of tokens. (Yu, 2022) highlights that in Llama-2-7B, the top 500 tokens out of 64,000 dominate the attention weights. Similarly, (Chen et al., 2024b) observes that the top 20% of tokens account for 80% of the attention weight, a concentration that occurs even more rapidly in the presence of attention sinks. By effectively and accurately identifying these important tokens, the number of tokens required for LLM inference can be significantly reduced without compromising the model’s performance.

2.3 LLM Long-context Inference

LLM long-context inference encounters two key challenges: system efficiency and task effectiveness.

LLMs struggle with system efficiency due to the high cost of memory storage and access. As the context length increases, the memory required for storing the KV cache scales linearly with the sequence length, eventually surpassing the available GPU memory capacity. For example, each decoding stage requires loading K and V values for self-attention, consuming up to 16GB of memory for a 32k context in Llama7b. In addition to limited GPU memory, the bottleneck lies in the memory access of KV cache, which takes up 53%(Tang et al., 2024) of the decoding time, underscoring the need to optimize self-attention for efficient long-context inference.

LLMs face limitations in task effectiveness, stemming from their lack of ability to capture long-distance dependencies and the challenge of forgetting information midway through processing. In the setting of 128K context, Llama3.1 (70B) can only get 66.6% score in the long-context benchmark RULER (Hsieh et al., 2024), indicating significant room for improvement.

2.4 Achilles’ Heel of Existing Methods

Static KV selection. Existing approaches like Streaming LLM (Xiao et al., 2023) and Snap KV(Li et al., 2024) rely on static selection or compression of the KV cache to address the memory constraints imposed by long-context inference. While these methods reduce memory usage, they often lack adaptability to dynamic input distributions and varying token importance. This rigidity can lead to a loss of critical contextual information, negatively impacting model performance on complex tasks

that require nuanced attention over long sequences.

Heuristic token selection. Methods such as H2O(Zhang et al., 2024) and Tidal-Decode(Yang et al., 2024) attempt to reduce computational overhead by heuristically selecting important tokens for processing. However, these techniques rely on fixed rules or predefined importance criteria, which may not generalize well across different tasks or input sequences. Consequently, important tokens may be overlooked, leading to suboptimal performance in scenarios requiring comprehensive contextual understanding.

Queries and keys similarity estimation. Approaches like Quest(Tang et al., 2024) use quantized dot-product similarity to estimate the relevance of keys to queries dynamically. Quest’s performance improvement in the needle-in-a-haystack task (achieving 65% in a 10k context setting compared to StreamingLLM’s 1%) underscores the critical role of dynamically estimating token importance. While this allows for a more flexible and adaptive mechanism to determine token importance, it still requires scanning all keys, leading to additional computational overhead.

Overall, these limitations underscore the need for more robust, efficient, and adaptable solutions to improve the scalability and effectiveness of attention mechanisms in long-context inference.

3 Methodology

Section 2.2 highlights the presence of sparsity in self-attention. Figures 1 and 2 illustrate the similarity of attention patterns and oracle tokens across continuous layers.

However, the above observation alone is insufficient to fully address the problem. A key challenge is that important tokens, particularly those in the middle of the sequence, gradually change, leading to a decline in the important token hit rate as layer distance increases. If we rely on the important token indices from one or a few layers for all layers, the hit rate cannot be maintained, as demonstrated in Figure 4.

To address this issue, we propose GRAPE, which enables sparse attention by incorporating heuristic information from two sources for token selection: (a) past decoding steps and (b) previous layers.

As shown in Figure 3, GRAPE aligns with two key stages of LLM inference: the prefill stage and

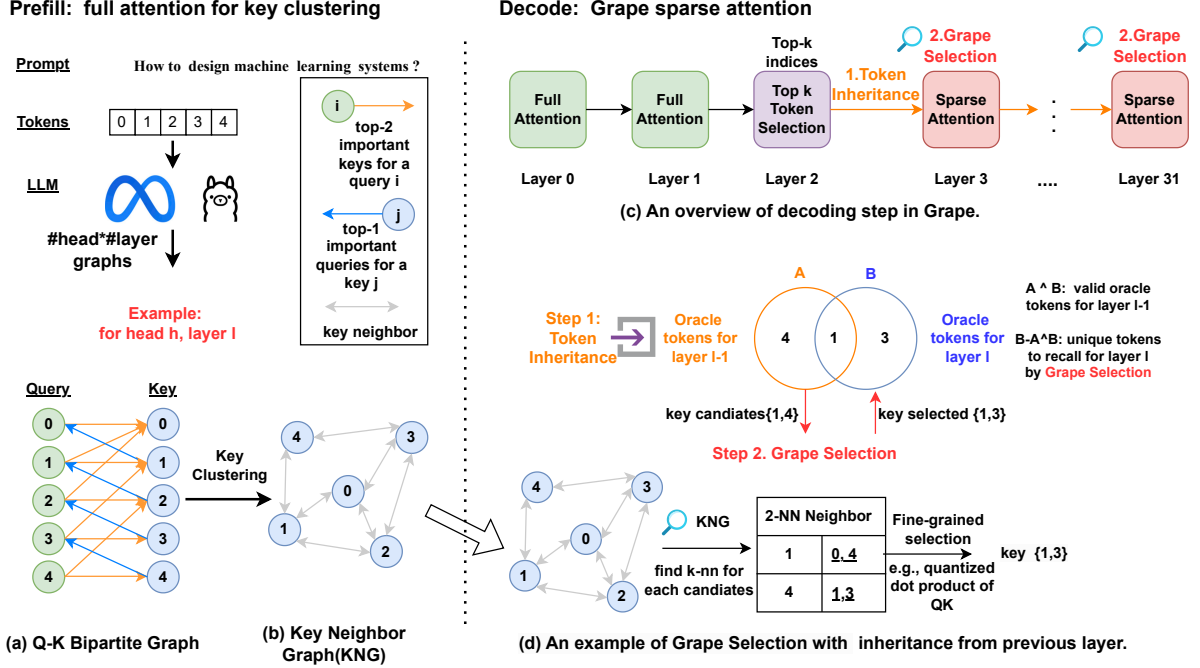


Figure 3: **GRAPE has two stages of procedure for sparse attention.** (Left) Prefill stage: full attention for key clustering with Key Neighbor Graph(KNG); (Right) Decode stage: Grape sparse attention with token inheritance and selection.

the decode stage. In the prefill stage, GRAPE collects heuristic information from prompt tokens to cluster tokens and keys into graphs. During the decode stage, GRAPE inherits important tokens and keys from the previous layer, expanding token candidates through heuristic graphs for selection. The two-stage design of GRAPE is detailed in Subsection 3.1 and 3.2. Algorithms for these two stages are shown in Algorithm 1 and Algorithm 2, respectively.

Algorithm 1 Prefill Stage: Key Clustering

Require: Prompt tokens’ queries Q , keys K , neighbor number K_n

Ensure: Key Neighbor Graphs (KNGs)

- 1: **for** each layer l and each attention head h **do**
- 2: Compute $D(Q, K) = Q \cdot K^\top$ {Dot product for attention similarity}
- 3: **for** each key k_j in K **do**
- 4: Find top-1 nearest query $q_{k_j}^{1NN}$.
- 5: Use $q_{k_j}^{1NN}$ ’s top- k_n similar (k_n -NN) keys of as neighbors for k_j .
- 6: **end for**
- 7: Build KNG for layer l , head h using k_j and its neighbors.
- 8: **end for**
- 9: **return** All KNGs

3.1 Prefill Stage: Key Clustering by Grape Graphs

The essence of token selection. Given a current query q_n and token budget b , top- k attention approximates the similarity between q_n and all keys $K[:,]$ by dot product $D(q_n, K) = q_n K^\top$, then select top- b indices (a.k.a, *oracle tokens* (Zhang et al., 2024)) $S = [s_1, s_2 \dots s_b]$, to get a subset of keys $K[S]$. We need to find a function $E(\cdot)$ whose overhead is less than $D(\cdot)$, to get samples of keys $S' = [s'_1, s'_2 \dots s'_b]$ that maximizes the the overlap $|S \cap S'|$. One potential option of $E(\cdot)$ is to adopt Approximate Nearest Neighbor Search (ANNS) for an incoming query q_i among the past queries, then make use of the heuristic information of top- b important tokens of past queries to get $K[S']$. For example, we can find 1-NN similar query of q_i in the past queries, which is noted as q_{1nn} , and adopt its top- k relevant keys to the q_i for attention calculation.

Why do we need to eliminate the storage of queries ANNS of queries to find the most relevant keys is inefficient in-memory storage, which introduces an additional memory footprint to store and access query vectors besides key vectors. We’d better eliminate the storage of past queries and only cluster the keys.

Algorithm 2 Decode Stage: Token Selection

Require: Previous layer tokens T_{L-1} , Key Neighbor Graphs (KNGs), query q_n , budget b , chunk size K , page size P

Ensure: Selected tokens T_L

- 1: **Step 1: Token Inheritance**
 - 2: $T_L \leftarrow T_{L-1}$ {Inherit tokens from the previous layer as pivots}
 - 3: **Step 2: GRAPE Selection**
 - 4: **for** each inherited token $t_i \in T_{L-1}$ **do**
 - 5: Use KNG to expand t_i into candidate keys.
 - 6: **end for**
 - 7: Divide candidate keys into pages of size P .
 {Each page contains P keys}
 - 8: **for** each page p **do**
 - 9: Compute page-level heuristic score:
 - 10: $H_p = \max(q_n \cdot K_p^{\min}, q_n \cdot K_p^{\max})$ {Where $K_p^{\min}$ and $K_p^{\max}$ are min and max keys in page p }
 - 11: **end for**
 - 12: Select top- b/P pages based on H_p . {Select top pages first}
 - 13: From selected pages, refine selection to obtain b tokens.
 - 14: **return** T_L
-

Why keys are enough for token selection? Recall that we find the similarity of important tokens between continuous layers, given a set of keys that contains the important keys for the current query, we only need to build ANNS of keys to expand key candidates to cover as many as the *oracle tokens*. See details of applying key clustering for token selection by Key Neighbor Graphs in Section 3.2.

How to cluster keys without storing queries in memory? The relationship between queries and keys can be regarded as a bipartite graph, as shown in Figure 3 (a). Inspired by RoarGraph(Chen et al., 2024a), we can project the relationship between queries and keys to only the internal of keys. In the left part of Figure 3, for each key vector k_i , we find 1-NN query $q_{k_i}^{1nn}$, then adopts 2-NN keys of $q_{k_i}^{1nn}$ as the neighbors of k_i . For each head in every layer, we will have a Key Neighbor Graph (KNG) because different heads focus on different tokens.

The cost of graph construction. The construction of KNGs happens in the prefill stage because we will get the dot product of all queries and keys for prompt tokens during the self-attention. The

cost of applying K-NN in those dot products of queries and keys is neglectable compared with the attention calculation itself. See Algorithm 1 for details.

3.2 Decode Stage: Token Inheritance and Selection

In Section 3.1, to extract the historical information among previous tokens, we introduce the intuition and details of key clustering in Key Neighbor Graphs(KNGs). In this subsection, we demonstrate how to use KNGs and information from previous layers for token selection.

Overview. In the right part of Figure 3, GRAPE incorporates three types of attention layers during each decoding step: full attention, full attention with top-k token selection, and GRAPE sparse attention. Initially, GRAPE applies full attention to the early Transformer layers to mitigate performance degradation, as noted in previous studies(Tang et al., 2024; Li et al., 2024). Next, the layer immediately following the full attention and a selected middle layer (e.g., layer 2 in Figure 3 (c)) utilize full attention with token selection. In this stage, GRAPE stores the inner product between the current query and key vectors of all tokens in the KV cache and selects the top-k tokens with the highest attention scores. Finally, the remaining layers (layer 3-31 in Llama 3-8B) utilize Grape sparse attention, which consists of two key steps: (1) token inheritance and (2) GRAPE selection, as detailed below. See Algorithm 2 for details.

Step 1:token inheritance. To calculate the Grape sparse attention in layer L, we inherit token indices selected from layer (L-1) as the pivot for key candidates expansion. Due to the high similarity (74-96% in Figure 2) of important tokens in two continuous layers, there would be some important tokens for layer L that could be inherited from layer (L-1) directly.

Step 2:Grape selection. However, the important tokens in the middle of the sequence change from layer to layer gradually. Therefore, it is important to recall the tokens that are unique for layer L. Grape adopts KNGs of layer L to expand more key candidates to increase the possibility of recalling important tokens. For example, in Figure 3 (d), token 1 and 4’s 2-NN neighbors contain the unique oracle token 3 of layer L. To satisfy the fixed token budget, after the coarse-grained selection before,

we adopt a quantized dot product (Tang et al., 2024) of the current query and key candidates for fine-grained selection. We group some tokens (e.g., per 4 tokens in the key candidates after coarse-grained selection) in a page, then keep track of the minimal and maximal Key values in KV cache pages and estimate the criticality of a given page using Query vectors.

Compared with Quest (Tang et al., 2024) that first introduces this technique, GRAPE has two advantages: (1) GRAPE only needs to scan the key candidates after coarse-grained selection, which takes less than 30% rather than all keys in Quest; (2) GRAPE can capture long distance better than Quest because the page of Quest contains continuous tokens in the sequence while the key candidates obtained by coarse-grained selection of GRAPE can be discrete in the sequence.

3.3 System Challenges and Co-Design

Expensive memory storage and memory access.

Recall that memory storage and access are expensive in LLM inference discussed in Section 2.3, one potential solution is to store KV cache both in GPU and CPU. GPU is fast for memory access and computation while the memory capacity is quite limited (e.g., NVIDIA A100 has < 80GB memory). In contrast, CPU-side memory provides ample memory capacity (e.g., 512 GB) with a bandwidth of 100–200 GB/s, typically only 10–20% of the bandwidth offered by GPU memory. The storage and loading cost of KNG can be ignored compared with the KV cache so we only deal with KV cache for high hardware utilization.

CPU-GPU Coordination. Therefore, we partition the KV cache to store either in CPU or GPU based on their heat for selection: (1) for the token that must be selected (such as the KV cache of the 0-1 layer), we keep the corresponding KV cache in the GPU; (2) for the KV cache of the selected token in the previous layer, we pre-fetch the KV cache to GPU to hide the offloading overhead; (3) If the KV cache of the remaining tokens to be selected cannot be stored in the GPU, they will be kept in the CPU.

4 Experiments

In this section, we aim to demonstrate that GRAPE can speed up LLM decoding while preserving high accuracy. We present GRAPE’s accuracy in downstream tasks, followed by our end-to-end system

results showing GRAPE’s performance.

4.1 Setting.

Testbed, Models, and Configurations. Our experiments are based on Llama3-8b-chat-128k (Grattafiori et al., 2024; Touvron et al., 2023) model with a single NVIDIA GTX A6000 GPU (48GB memory).

Benchmark. Our experiments include 13 synthetic tasks from RULER (Hsieh et al., 2024) with 100 examples per task. The context length is 4k. RULER comprises tasks across four categories: *retrieval*, *multi-hop tracing*, *aggregation*, and *question answering*. We list the main tasks in each category below.

- *Retrieval*: Needle-in-a-haystack (NIAH) test has been commonly employed to evaluate long-context modeling capabilities of language models. RULER extended the vanilla NIAH test to different forms including **single NIAH** (S-NIAH), **multi-keys NIAH** (MK-NIAH), **multi-values NIAH** (MV-NIAH), and **multi-queries NIAH** (MQ-NIAH).
- *Multi-hop Tracing*: RULER develops a **variable tracking** (VT) task to emulate a minimal coreference chain solution task, which checks the behavior of checking relevant co-occurrence patterns and drawing skipped connections.
- *Aggregation*: RULER also introduces **common words extraction** (CWE) and **frequent words extraction** (FWE) tasks as a proxy for summarization tasks where relevant information constitutes much larger portion and the target output depends on accurate aggregation.
- *Question Answering*: RULER inserts golden paragraphs into randomly sampled paragraphs from the existing **question answering** (QA) dataset to design the task of QA.

Baselines. In our experiments, we use SOTA-Quest (Tang et al., 2024) as our major baseline. We evaluate Quest with a KV page size of 8. In the meantime, we also use **position-based token selection** (POS) as our baselines. POS adopts *Oracle Tokens* from the second layer for further layers and resamples at 5,10,15,25 layers for better performance in end-to-end accuracy. As references, we also give the results of full-attention and top-*k*

token selection, which serve as upper bounds without/with token budget limitations for the accuracy performance across different tasks. For all baselines, the initial 4 tokens and the latest 16 local tokens as well as layer-0, 1 are statically preserved.

4.2 Effectiveness of GRAPE

In NIAH tasks with 4K context length, we first verify the effectiveness of GRAPE in (1) *Oracle Token* hit rate, and (2) attention similarity as follows. We compare GRAPE with position-based token selection (*POS*) which adopts oracle tokens from layer 2 to the layers afterward.

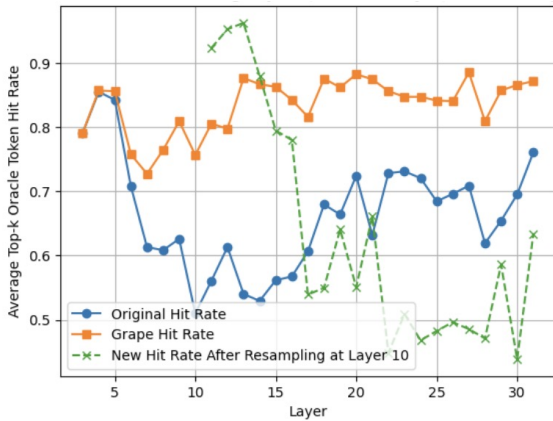


Figure 4: Top-32 important (a.k.a, oracle) token hit rate of different sparse attention in Llama3-8B. *Original Hit Rate* (in blue) is for *POS*, and *New Hit Rate After Resampling at Layer 10* (in green) is to resample at layer 10 for tokens applied to 11-31 layers. *Grape Hit Rate* (in orange) is significantly better than others.

Oracle Token hit rate. We get the top-32 important tokens of each layer as oracle tokens, and then obtain the selected tokens of different methods for oracle token hit rate. As shown in Figure 4, *Grape Hit Rate* (in orange) is significantly better than others. For *POS* (in blue), the hit rate decays when the distance between the current layer and layer 2 increases. Even resampling at layer 10 (in green) cannot solve the performance degradation for 19-31 layers.

Attention similarity. We calculate the cosine similarity between different sparse attention and full attention distributions. Top-k Attention (in blue) is the upper bound for a given token budget. *POS* (in orange) shows 0.88 in layer 10, while *GRAPE* (in green) can improve the cosine similarity to 0.95 in layer 10.

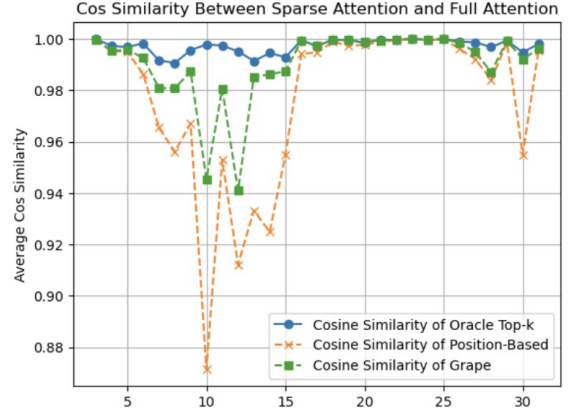


Figure 5: The Cosine similarity between different sparse attention and full attention in Llama3-8B. The token budget for sparse attention (*Top-k* in blue, *POS* in orange, and *GRAPE* in green) is 32.

4.3 End-to-end Accuracy

We test GRAPE and different sparse attention baselines in 13 tasks of RULER for end-to-end accuracy. The context length is about 4k tokens and the model is Llama3-8b-chat-128k. The token budget includes 16, 32 and 64. We only show the result for a budget of 16 tokens in this subsection and delay the discussion of different token budgets in Section 4.4. We also show the result of full attention and top-k attention, which adopts all tokens/top-k oracle tokens as the performance upper bounds.

We demonstrate the accuracy of different methods in Table 1. From Table 1, we observe that GRAPE preserves higher accuracy in 12 out of 13 tasks than other baselines including POS and Quest. For S-NIAH-1, S-NIAH-2, QA-1 and QA-2, GRAPE even reaches the upper bound performance of Top-k Attention. For some tasks where Quest and POS’s accuracy is lower than 50%, including MK-NIAH-2, MQ-NIAH, VT, and FWE, GRAPE can have up to 40% higher accuracy than SOTA-Quest. We hypothesis that those tasks have important information distributed over a long distance and GRAPE can identify those tokens well with the help of KNG.

4.4 Sensitivity To Token Budget

The sensitivity analysis of GRAPE across varying token budgets (16, 32, 64) highlights its adaptability and effectiveness in long-context tasks. GRAPE consistently outperforms baseline methods such as Quest and POS in the majority of tasks, demonstrating significant improvements in average accuracy with limited token budgets. For the token budget

Methods	Token bud- get	S- NIAH- 1	S- NIAH- 2	S- NIAH- 3	MK- NIAH- 1	MK- NIAH- 2	MK- NIAH- 3	MV- NIAH	MQ- NIAH	VT	CWE	FWE	QA- 1	QA- 2	Avg.
Full Attention		100.00	100.00	100.00	100.00	100.00	99.00	99.25	99.75	99.80	99.90	93.00	92.00	66.00	96.05
Top-K	b=16	100.00	100.00	100.00	100.00	100.00	97.00	96.75	98.25	97.20	56.80	80.33	93.00	64.00	91.03
POS	b=16	95.00	87.00	61.00	87.00	16.00	2.00	46.50	55.00	15.60	6.50	29.67	87.00	56.00	49.56
Quest	b=16	99.00	97.00	38.00	92.00	47.00	0.00	51.75	73.50	55.80	18.50	35.00	88.00	61.00	58.20
GRAPE	b=16	100.00	100.00	40.00	99.00	89.00	10.00	82.25	87.50	94.40	39.10	69.00	93.00	64.00	74.40

Table 1: GRAPE outperforms POS and Quest in 12 out of 13 tasks with a token budget of 16. The context length is 4K. Full Attention and Top-k Attention are the performance upper bounds without/with a token budget limitation, respectively. The best sparse attention in a certain task is in **bold**.

Methods	Token Bud- get	S- NIAH- 1	S- NIAH- 2	S- NIAH- 3	MK- NIAH- 1	MK- NIAH- 2	MK- NIAH- 3	MV- NIAH	MQ- NIAH	VT	CWE	FWE	QA- 1	QA- 2	Avg.
Quest	b=16	99.00	97.00	38.00	92.00	47.00	0.00	51.75	73.50	55.80	18.50	35.00	88.00	61.00	58.20
	b=32	99.00	100.00	97.00	100.00	77.00	2.00	87.25	96.00	88.40	29.40	63.00	91.00	62.00	76.31
	b=64	99.00	100.00	99.00	100.00	93.00	15.00	97.75	99.75	98.40	46.00	76.00	93.00	62.00	82.99
POS	b=16	95.00	87.00	61.00	87.00	16.00	2.00	46.50	55.00	15.60	6.50	29.67	87.00	56.00	49.56
	b=32	94.00	92.00	67.00	89.00	22.00	2.00	49.25	62.00	19.80	7.80	34.67	88.00	55.00	52.50
	b=64	97.00	91.00	72.00	90.00	26.00	10.00	53.00	63.50	22.80	12.20	56.00	88.00	59.00	56.96
GRAPE	b=16	100.00	100.00	40.00	99.00	89.00	10.00	82.25	87.50	94.40	39.10	69.00	93.00	64.00	74.40
	b=32	100.00	100.00	39.00	100.00	78.00	1.00	84.50	91.00	92.60	43.40	67.33	93.00	65.00	73.45
	b=64	100.00	100.00	44.00	100.00	94.00	35.00	93.25	95.50	98.00	56.70	71.67	93.00	65.00	80.47

Table 2: GRAPE’s performance across different tasks given different token budgets including 16, 32, and 64. The best sparse attention for each budget setting in a certain task is in **bold**. For the token budget size of 16/32/64, GRAPE outperforms others in 12/9/8 out of 13 tasks in RULER, respectively.

size of 16/32/64, GRAPE outperforms others in 12/9/8 out of 13 tasks in RULER, respectively. Detailed results compared with baseline conditions are demonstrated in Table 2.

GRAPE’s ability to select tokens from long distances enables it to perform better in tasks requiring a broad contextual understanding, such as MQ-NIAH and VT. Its integration of historical and vertical information ensures it retains critical information across layers, as evidenced by its higher accuracy compared to Quest and POS in these scenarios.

While GRAPE excels in most tasks, its performance on S-NIAH-3 slightly lowers the overall average accuracy. This task requires identifying UUIDs (32 digits) within a long context, where Quest has an advantage due to its continuous paging mechanism. GRAPE, on the other hand, selects tokens based on their importance, including those at longer distances. Under smaller token budgets, GRAPE may miss some consecutive tokens needed to fully reconstruct the answer. For example, GRAPE might correctly select 27 out of 32 UUID tokens, but their non-contiguity prevents successful task completion. This limitation is inherent to GRAPE’s design, as it prioritizes critical token diversity over continuity.

4.5 Discussion of Efficiency&Future Plan

Recall Section 4.3, GRAPE achieves high accuracy with a token budget of 16 in a 4k context for most RULER tasks, reducing memory usage by $250\times$.

For end-to-end latency, GRAPE achieves a $1.3\times$ speedup over full attention in MK-NIAH-2. However, further optimization is possible by: (1) evaluating GRAPE on longer contexts ($>16k$ tokens in Llama3-8B), where attention overhead dominates latency, and (2) implementing a CUDA kernel for GRAPE attention to ensure a fair comparison, as full attention leverages optimized Flash Attention in CUDA.

5 Conclusion

We present GRAPE, a graph-based KV cache selection algorithm for efficient long-context LLM inference. GRAPE combines historical and vertical information through Key Neighbor Graphs (KNGs), inheriting tokens from previous layers and expanding candidates using heuristic graphs. On the RULER dataset, GRAPE achieves 74.40% task accuracy with a token budget of 16, significantly outperforming the 58.20% accuracy of Quest. GRAPE reduces memory overhead while maintaining high accuracy, offering a scalable and efficient solution for long-context LLMs.

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